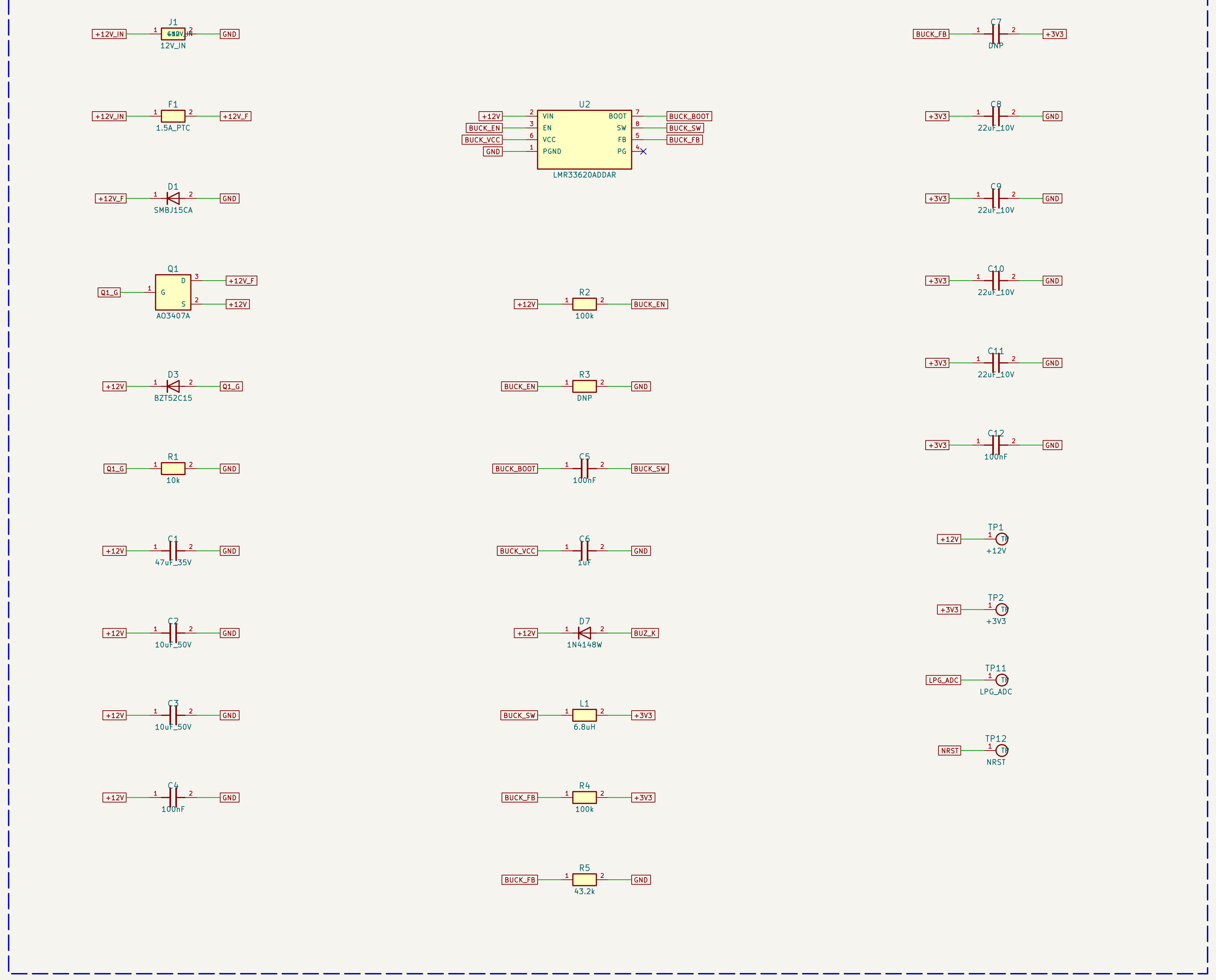
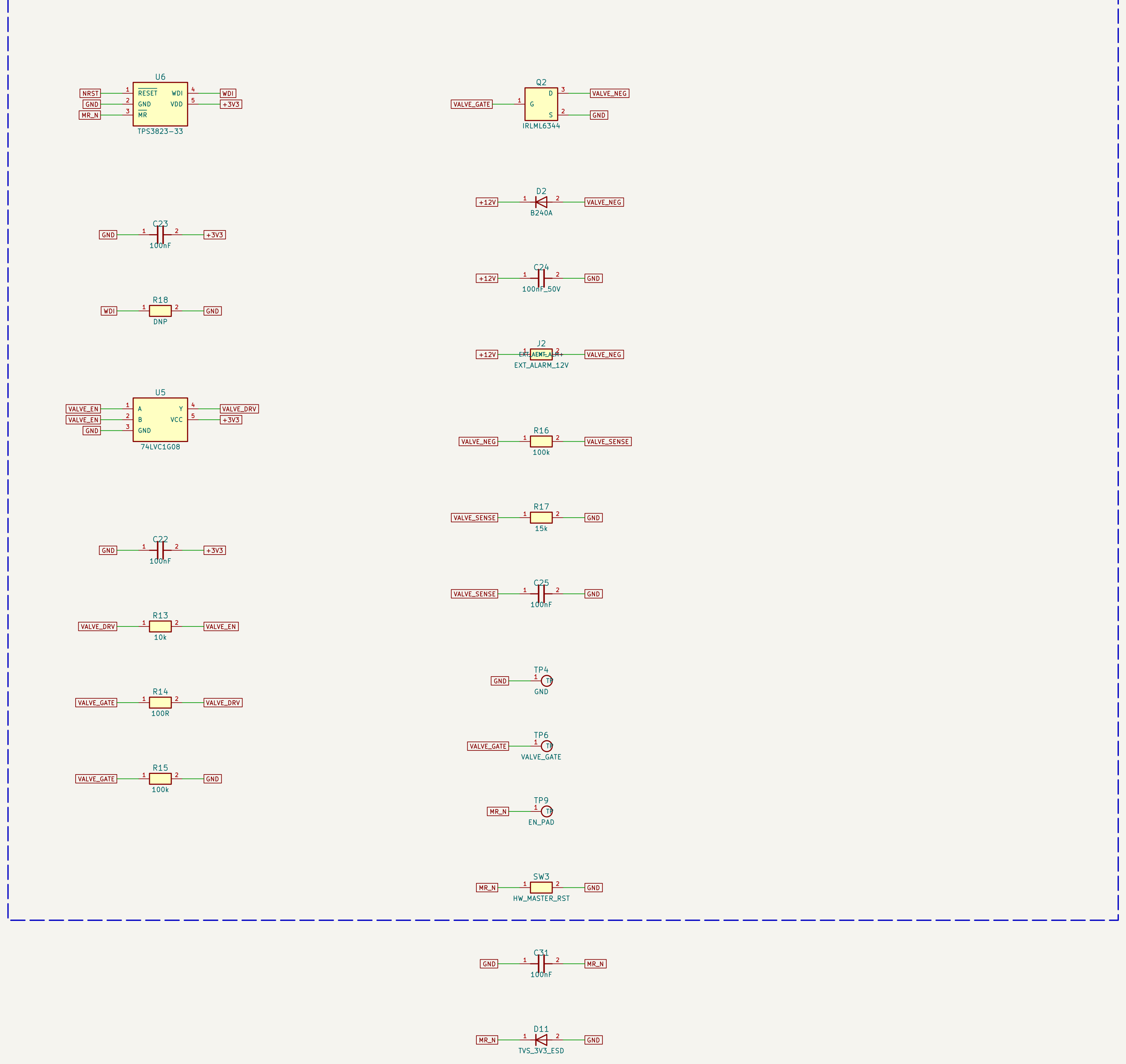
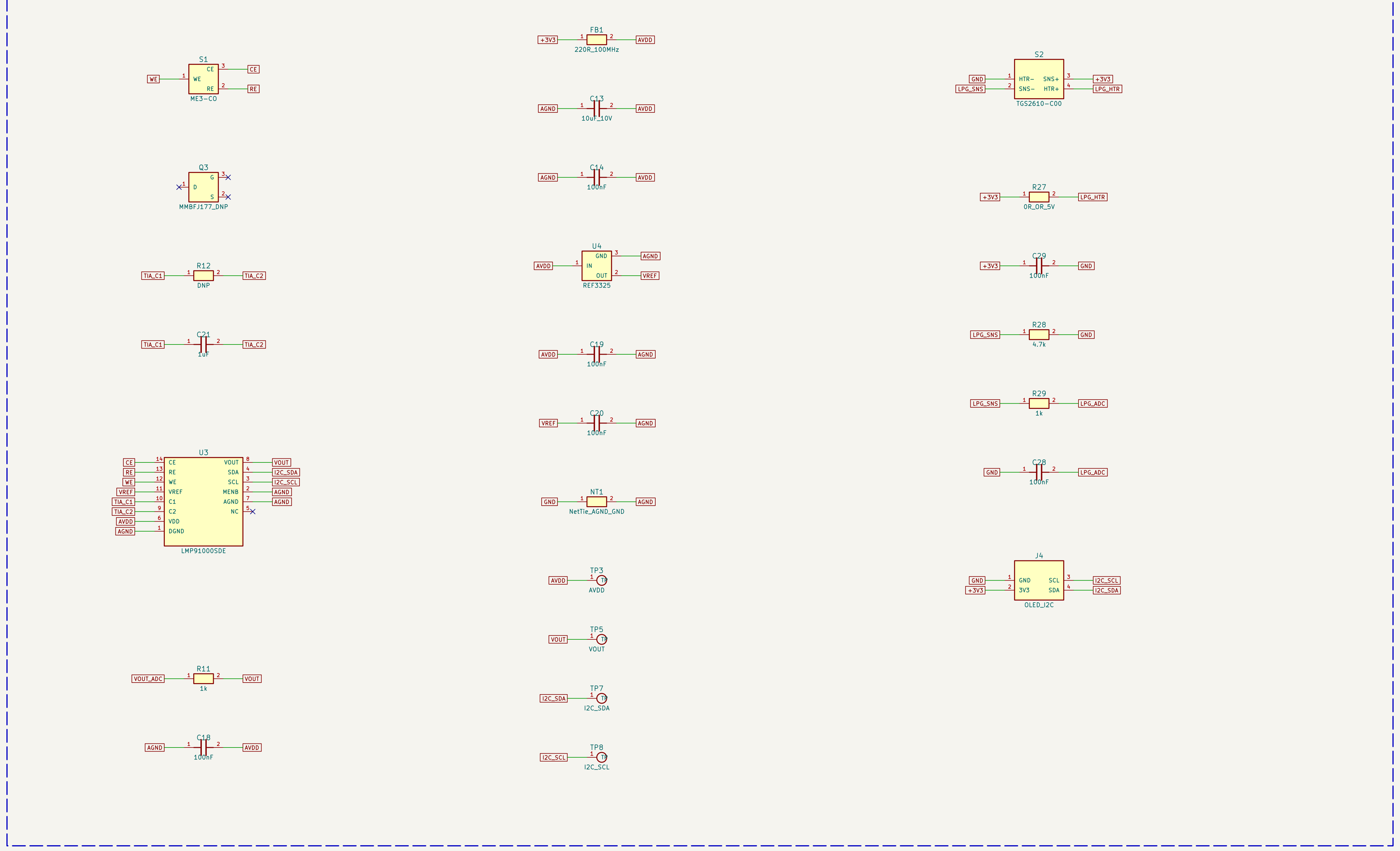
BLOCK 1: 12V INPUT PROTECTION & 3.3V 2A SYNCHRONOUS BUCK CONVERTER (LMR33620ADDAR)



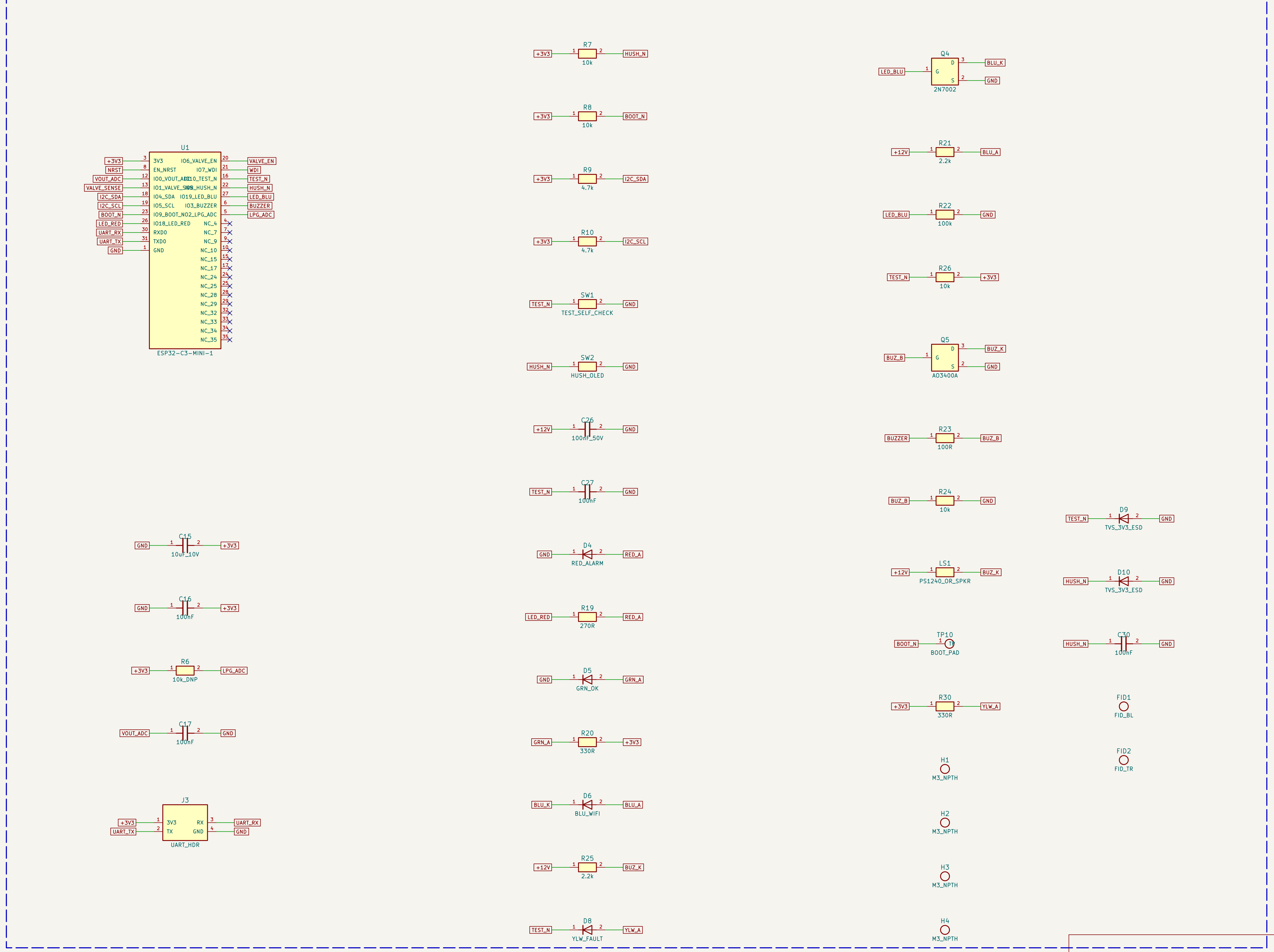
BLOCK 2: HARDWARE WATCHDOG (TPS3823-33), SAFETY INTERLOCK (74LVC1G08) & 12V EXTERNAL ALARM OUTPUT (J2)



BLOCK 3: DUAL GAS SENSORS – CO (ME3-CO + LMP91000 AFE) & LPG (FIGARO TGS2610-C00) + I2C OLED (J4)



BLOCK 4: ESP32-C3-MINI-1 WIRELESS MCU, DYNAMIC AUDIO SPEAKER (CDM-12008), LEDS, BUTTONS & UART



Power: 12V Protected Input + LMR33620ADDAR 3.3V Buck | Safety: TPS3823-33 + 74LVC1G08 + IRLM
MCU: ESP32-C3-MINI-1 | AFE: LMP91000SDE + REF3325 + Winsen ME3-CO
Antigravity Hardware Engineering
Sheet: /
File: co_detector_controller.kicad_sch
Title: CO Detector & Automatic 12V Gas Shut-Off Valve Controller
Size: A1
Date: 2026-11-07
Rev: 1.0
KiCad E.D.A. 10.0.6
Id: 1/1